

COMPREHENSIVE FSD PROMPT – Super Admin, Admin & Vendor App
Complete End-to-End Specification with Technical Flows

PART 0: CONTEXT & ANALYSIS

Current Status

You are analyzing and building upon an **EXISTING** React Native + Expo application that already has:

- Login page (functional)
- Initial loading page (splash/startup flow)
- Partial vendor dashboard (incomplete)

Mission

Complete, fix, and implement the **entire ecosystem** for:

1. **Super Admin (Web/Django)**
2. **Admin (Web/Django)**
3. **Vendor App (React Native/Expo + Django API)**

Focus: **Venues/Courts, Coaches, Ecommerce** management.

Tools & Stack

- **Backend:** Django 5.x + Django REST Framework + PostgreSQL
- **Frontend (Web):** Django templates (minimal) or React (optional admin dashboard)
- **Frontend (Mobile):** React Native + Expo + TypeScript
- **Payment:** Razorpay
- **Storage:** Firebase/S3
- **Auth:** JWT + 2FA
- **Design:** Modern, Material Design 3 (Expo), responsive, dark/light mode

PART 1: DATABASE SCHEMA & MODELS

1.1 User Management Models

UserRole Enum

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SUPER_ADMIN = 'super_admin'

ADMIN = 'admin'

VENDOR = 'vendor'

CUSTOMER = 'customer'

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User Model (Django)

Fields:

- `id` (UUID, primary key)
- `email` (unique, required, validated)
- `phone_number` (unique, nullable, E.164 format)
- `password` (hashed, bcrypt minimum 12 rounds)
- `first_name` (string, 1-100 chars)

- `last\_name` (string, 1-100 chars)
- `avatar\_url` (file URL, nullable)
- `role` (enum: SUPER_ADMIN, ADMIN, VENDOR, CUSTOMER)
- `is\_active` (boolean, default True)
- `is\_verified` (boolean, email/phone verified, default False)
- `is\_2fa\_enabled` (boolean, default False)
- `2fa\_secret` (encrypted string, nullable)
- `created\_at` (timestamp, auto)
- `updated\_at` (timestamp, auto)
- `last\_login` (timestamp, nullable)
- `login\_attempts` (integer, default 0, reset after success)
- `locked\_until` (timestamp, nullable, after 5 failed attempts, 30 min lock)
- `created\_by` (FK to User, nullable, audit)
- `updated\_by` (FK to User, nullable, audit)
- `metadata` (JSON, device info, preferences, nullable)

****Validations:****

- Email regex: RFC 5322 compliant
- Phone: E.164 format (+91XXXXXXXXXX for India)
- Password: min 12 chars, 1 uppercase, 1 lowercase, 1 digit, 1 special char

AdminHierarchy Model (for Admin under Super Admin)

****Fields:****

- `id` (UUID)
- `admin\_user` (FK to User, role=ADMIN)
- `super\_admin\_user` (FK to User, role=SUPER_ADMIN)
- `vendor\_access` (array of vendor IDs, allowed to manage, or 'ALL')
- `permissions` (JSON: can_onboard_vendor, can_manage_customers, can_view_analytics, can_moderate_offers)
- `created\_at` (timestamp)
- `assigned\_at` (timestamp, who assigned)

1.2 Vendor & Service Models

Vendor Model

****Fields:****

- `id` (UUID)
- `user` (FK to User, role=VENDOR)
- `vendor\_type` (enum: VENUE, COACH, ECOMMERCE, HYBRID)
- `business\_name` (string, 1-200 chars)
- `legal\_name` (string, required)
- `description` (text, optional, 0-5000 chars)
- `logo\_url` (file URL)
- `banner\_url` (file URL)
- `email` (string, validated)
- `phone\_number` (E.164)
- `website` (URL, optional)
- `address` (string, 1-500 chars)
- `city` (string, required)
- `state` (string, required)

- `postal\_code` (string, required)
- `country` (string, default 'IN')
- `latitude` (decimal, 8 places)
- `longitude` (decimal, 8 places)
- `kyc\_status` (enum: PENDING, VERIFIED, REJECTED, default PENDING)
- `kyc\_document\_url` (file URL)
- `kyc\_verified\_at` (timestamp, nullable)
- `kyc\_verified\_by` (FK to User/Admin, nullable)
- `pan\_number` (string, 10 chars, required, PAN validation regex)
- `gst\_in` (string, 15 chars, optional, GSTIN validation)
- `bank\_account\_holder` (string)
- `bank\_name` (string)
- `account\_number` (encrypted string)
- `ifsc\_code` (string, 11 chars)
- `account\_type` (enum: SAVINGS, CURRENT)
- `upi\_id` (string, optional, for fast payouts)
- `payout\_frequency` (enum: DAILY, WEEKLY, MONTHLY, default WEEKLY)
- `commission\_rate` (decimal, 5,2, default 20.00 %)
- `custom\_commission` (boolean, if Super Admin allows per-vendor override)
- `approval\_status` (enum: PENDING, APPROVED, REJECTED, SUSPENDED)
- `approved\_at` (timestamp, nullable)
- `approved\_by` (FK to Admin, nullable)
- `onboarded\_at` (timestamp)
- `onboarded\_by` (FK to Admin/Super Admin)
- `is\_active` (boolean, default True)
- `suspension\_reason` (text, nullable)
- `staff\_count` (integer, count of team members)
- `total\_bookings` (integer, analytics, denormalized)
- `total\_revenue` (decimal, 10,2, analytics, denormalized)
- `avg\_rating` (decimal, 2,1, nullable)
- `total\_ratings` (integer, default 0)
- `created\_at` (timestamp)
- `updated\_at` (timestamp)
- `deleted\_at` (timestamp, soft delete, nullable)

****Indexes:****

- vendor_type, approval_status, city, created_at (for queries)
- user_id (unique FK)

VendorStaff Model (Vendor's Internal Staff)

****Fields:****

- `id` (UUID)
- `vendor` (FK to Vendor)
- `user` (FK to User)
- `role` (enum: OWNER, MANAGER, STAFF, ACCOUNTANT)
- `permissions` (JSON: can_create_bookings, can_edit_pricing, can_view_reports, can_manage_staff)
- `assigned\_at` (timestamp)
- `assigned\_by` (FK to User)
- `is\_active` (boolean)
- `created\_at` (timestamp)

1.3 Venue/Court Models (for Venue Vendors)

Venue Model

****Fields:****

- `id` (UUID)
- `vendor` (FK to Vendor)
- `name` (string, 1-200 chars)
- `description` (text, optional)
- `address` (string)
- `latitude` (decimal)
- `longitude` (decimal)
- `phone\_number` (E.164)
- `images` (JSON array: [url1, url2, ...], max 10)
- `amenities` (JSON array: ['WiFi', 'Parking', 'Cafeteria', ...])
- `opening\_time` (time, HH:MM)
- `closing\_time` (time, HH:MM)
- `is\_open\_24\_7` (boolean)
- `holiday\_dates` (JSON array: ['2025-12-25', '2025-01-26', ...])
- `cancellation\_window\_hours` (integer, default 24)
- `reschedule\_window\_hours` (integer, default 48)
- `is\_verified` (boolean, admin verified)
- `verified\_at` (timestamp, nullable)
- `verified\_by` (FK to Admin, nullable)
- `is\_active` (boolean)
- `created\_at` (timestamp)
- `updated\_at` (timestamp)

Court Model

****Fields:****

- `id` (UUID)
- `venue` (FK to Venue)
- `name` (string, 1-100 chars)
- `court\_type` (enum: CRICKET, BADMINTON, TENNIS, BASKETBALL, VOLLEYBALL, FOOTBALL, TABLE_TENNIS, SWIMMING, GYM, YOGA, OTHER)
- `base\_price\_per\_hour` (decimal, 8,2, required)
- `currency` (string, default 'INR')
- `surface\_type` (string, optional)
- `capacity` (integer, max players)
- `lights\_available` (boolean)
- `equipment\_available` (JSON array)
- `images` (JSON array, max 10)
- `is\_active` (boolean)
- `created\_at` (timestamp)
- `updated\_at` (timestamp)

CourtAvailability Model

****Fields:****

- `id` (UUID)
- `court` (FK to Court)
- `day\_of\_week` (enum: 0-6, Monday-Sunday)
- `start\_time` (time)

- `end\_time` (time)
- `is\_available` (boolean)

TimeSlot Model (for bookable slots)

****Fields:****

- `id` (UUID)
- `court` (FK to Court)
- `date` (date)
- `start\_time` (time)
- `end\_time` (time)
- `duration\_minutes` (integer)
- `base\_price` (decimal, 8,2)
- `status` (enum: AVAILABLE, BOOKED, BLOCKED, MAINTENANCE)
- `booked\_by` (FK to Booking, nullable)
- `created\_at` (timestamp)
- `updated\_at` (timestamp)

1.4 Coach Models

Coach Model

****Fields:****

- `id` (UUID)
- `vendor` (FK to Vendor)
- `user` (FK to User)
- `first\_name` (string)
- `last\_name` (string)
- `bio` (text, optional, max 1000)
- `profile\_image` (file URL)
- `specializations` (JSON array: ['Cricket - Batting', 'Fitness', ...])
- `years\_of\_experience` (integer)
- `certifications` (JSON array: [{name, issuer, number, doc_url}, ...])
- `hourly\_rate` (decimal, 8,2)
- `is\_verified` (boolean, admin verified)
- `verification\_status` (enum: PENDING, VERIFIED, REJECTED)
- `verified\_at` (timestamp, nullable)
- `verified\_by` (FK to Admin, nullable)
- `available\_for\_online` (boolean)
- `available\_for\_venue` (boolean)
- `available\_for\_home` (boolean)
- `max\_home\_travel\_km` (integer, nullable)
- `home\_visit\_extra\_charge` (decimal, 8,2, nullable)
- `max\_clients\_per\_session` (integer, default 1)
- `cancellation\_policy` (text)
- `avg\_rating` (decimal, 2,1)
- `total\_ratings` (integer)
- `total\_sessions` (integer, denormalized)
- `is\_active` (boolean)
- `created\_at` (timestamp)
- `updated\_at` (timestamp)

CoachAvailability Model

****Fields:****

- `id` (UUID)
- `coach` (FK to Coach)
- `day\_of\_week` (enum: 0-6)
- `start\_time` (time)
- `end\_time` (time)
- `recurring` (boolean, weekly repeat)
- `lead\_time\_hours` (integer, min hours before booking)

CoachPackage Model (for bundled sessions)

****Fields:****

- `id` (UUID)
- `coach` (FK to Coach)
- `name` (string, e.g., '10-Session Package')
- `description` (text)
- `sessions\_count` (integer)
- `total\_price` (decimal, 8,2)
- `validity\_days` (integer, e.g., 90)
- `discount\_percentage` (decimal, 3,1)
- `is\_active` (boolean)

1.5 Ecommerce Models

EcommerceVendor Model

****Fields:****

- `id` (UUID)
- `vendor` (FK to Vendor)
- `shop\_name` (string)
- `shop\_description` (text)
- `logo\_url` (file URL)
- `banner\_url` (file URL)
- `categories` (JSON array: ['Equipment', 'Apparel', 'Nutrition', ...])

Product Model

****Fields:****

- `id` (UUID)
- `vendor` (FK to Vendor / EcommerceVendor)
- `sku` (string, unique per vendor)
- `name` (string, 1-200 chars)
- `description` (text)
- `category` (enum: EQUIPMENT, APPAREL, FOOTWEAR, ACCESSORIES, NUTRITION, DIGITAL)
- `price` (decimal, 8,2)
- `cost\_price` (decimal, 8,2, internal)
- `discount\_percentage` (decimal, 3,1, optional)
- `currency` (string, default 'INR')
- `images` (JSON array, max 10)
- `stock\_quantity` (integer)
- `low\_stock\_threshold` (integer, default 10)
- `sku` (string, unique)

- `weight\_kg` (decimal, optional)
- `dimensions\_cm` (JSON: {length, width, height}, optional)
- `shipping\_class` (enum: REGULAR, OVERSIZE, FRAGILE)
- `is\_returnable` (boolean)
- `return\_window\_days` (integer, default 30)
- `warranty\_description` (text, optional)
- `is\_active` (boolean)
- `created\_at` (timestamp)
- `updated\_at` (timestamp)

ProductVariant Model (for size/color)

****Fields:****

- `id` (UUID)
- `product` (FK to Product)
- `variant\_name` (string, e.g., 'Size: L, Color: Blue')
- `sku\_variant` (string)
- `price\_modifier` (decimal, can be +/-)
- `stock\_quantity` (integer)

1.6 Booking & Order Models

Booking Model (Venues/Courts)

****Fields:****

- `id` (UUID)
- `court` (FK to Court)
- `customer\_user` (FK to User, role=CUSTOMER)
- `customer\_name` (string)
- `customer\_phone` (E.164)
- `customer\_email` (string)
- `date` (date)
- `start\_time` (time)
- `end\_time` (time)
- `duration\_minutes` (integer)
- `base\_price` (decimal, 8,2)
- `discount\_amount` (decimal, 8,2, nullable, from offers)
- `applied\_offer\_id` (FK to Offer, nullable)
- `taxable\_amount` (decimal, 8,2)
- `tax\_percentage` (decimal, 3,1, default 18.0)
- `tax\_amount` (decimal, 8,2)
- `commission\_percentage` (decimal, 5,2)
- `commission\_amount` (decimal, 8,2)
- `total\_amount` (decimal, 8,2)
- `vendor\_payout` (decimal, 8,2)
- `payment\_method` (enum: ONLINE, CASH, WALLET, UPI)
- `payment\_status` (enum: PENDING, PAID, FAILED, REFUNDED)
- `payment\_id` (string, Razorpay ID, nullable)
- `booking\_status` (enum: CONFIRMED, COMPLETED, CANCELLED, RESCHEDULED)
- `cancellation\_reason` (text, nullable)
- `refund\_amount` (decimal, 8,2, nullable)
- `refund\_status` (enum: PENDING, PROCESSED, FAILED, nullable)
- `internal\_notes` (text, vendor notes, not visible to customer)

- `customer\_notes` (text, customer special requests)
- `created\_at` (timestamp)
- `updated\_at` (timestamp)
- `created\_by` (FK to User, vendor staff or admin)

CoachingSession Model

****Fields:****

- Similar to Booking, but for coaching:
- `id` (UUID)
- `coach` (FK to Coach)
- `customer\_user` (FK to User)
- `session\_type` (string, e.g., '1-on-1 Cricket Batting')
- `date` (date)
- `start\_time` (time)
- `end\_time` (time)
- `delivery\_mode` (enum: ONLINE, VENUE, HOME, HYBRID)
- `meeting\_link` (string, Zoom/Teams URL if online)
- `location\_address` (string, if home or venue)
- `base\_price` (decimal, 8,2)
- `[...same payment/offer/commission fields as Booking...]
- `attendance\_status` (enum: NOT_STARTED, IN_PROGRESS, COMPLETED, NO_SHOW, CANCELLED)
- `feedback\_rating` (decimal, 1-5, nullable)
- `feedback\_comment` (text, nullable)

Order Model (Ecommerce)

****Fields:****

- `id` (UUID)
- `vendor` (FK to Vendor)
- `customer\_user` (FK to User)
- `order\_number` (string, unique, auto-generated)
- `items` (JSON array: [{product_id, variant_id, quantity, price}, ...])
- `subtotal` (decimal, 8,2)
- `discount\_amount` (decimal, 8,2, from offers)
- `applied\_offer\_id` (FK to Offer, nullable)
- `shipping\_cost` (decimal, 8,2)
- `tax\_amount` (decimal, 8,2)
- `total\_amount` (decimal, 8,2)
- `commission\_percentage` (decimal, 5,2)
- `commission\_amount` (decimal, 8,2)
- `vendor\_payout` (decimal, 8,2)
- `payment\_method` (enum: ONLINE, COD, WALLET, UPI)
- `payment\_status` (enum: PENDING, PAID, FAILED, REFUNDED)
- `payment\_id` (string, Razorpay ID)
- `order\_status` (enum: PENDING, CONFIRMED, PROCESSING, SHIPPED, DELIVERED, CANCELLED, RETURNED)
- `shipping\_address` (JSON: {name, phone, address, city, state, postal_code})
- `tracking\_number` (string, nullable)
- `tracking\_url` (string, nullable)
- `courier\_name` (string, nullable)
- `fulfillment\_status` (enum: PENDING, SHIPPED, DELIVERED)
- `refund\_amount` (decimal, nullable)

- `refund\_status` (enum: PENDING, PROCESSED, FAILED, nullable)
- `created\_at` (timestamp)
- `updated\_at` (timestamp)

1.7 Offer & Commission Models

Offer Model

****Fields:****

- `id` (UUID)
- `created\_by` (FK to User, SUPER_ADMIN or ADMIN)
- `offer\_type` (enum: INITIAL, FESTIVE, REFERRAL, GENERAL, CUSTOM)
- `name` (string, e.g., 'Welcome 20% Off')
- `description` (text)
- `discount\_type` (enum: PERCENTAGE, FLAT_AMOUNT, BUY_X_GET_Y, FREE_SHIPPING)
- `discount\_value` (decimal, 8,2)
- `max\_discount\_cap` (decimal, 8,2, nullable, for % discounts)
- `minimum\_booking\_amount` (decimal, 8,2, optional)
- `applicable\_categories` (JSON array: ['VENUE', 'COACH', 'ECOMMERCE'])
- `applicable\_vendor\_ids` (JSON array, or NULL for all)
- `start\_date` (date)
- `end\_date` (date)
- `start\_time` (time, optional, specific hour)
- `end\_time` (time, optional)
- `applicable\_days\_of\_week` (JSON array: [0, 1, 2, 3, 4, 5, 6])
- `usage\_limit\_total` (integer, total redemptions allowed)
- `usage\_limit\_per\_customer` (integer, per person max uses)
- `exclude\_customer\_ids` (JSON array, optional)
- `include\_customer\_ids` (JSON array, optional)
- `is\_active` (boolean)
- `visibility\_percentage` (integer, 10/20/30, "show only top X%")
- `redemption\_count` (integer, current redemptions, denormalized)
- `revenue\_impact` (decimal, 10,2, analytics, denormalized)
- `created\_at` (timestamp)
- `updated\_at` (timestamp)
- `deleted\_at` (timestamp, soft delete)

CommissionConfig Model

****Fields:****

- `id` (UUID)
- `super\_admin` (FK to User)
- `default\_commission\_percentage` (decimal, 5,2, default 20.00)
- `category\_overrides` (JSON: {'VENUE': 18.0, 'COACH': 22.0, ...})
- `vendor\_custom\_rates` (JSON: {vendor_id: rate, ...})
- `payout\_cycle` (enum: DAILY, WEEKLY, MONTHLY)
- `minimum\_payout\_threshold` (decimal, 8,2)
- `tax\_rate\_gst` (decimal, 3,1, default 18.0)
- `is\_active` (boolean)
- `created\_at` (timestamp)
- `updated\_at` (timestamp)

1.8 Membership Models

Membership Model

****Fields:****

- `id` (UUID)
- `vendor` (FK to Vendor)
- `name` (string, e.g., 'Monthly Court Pass')
- `description` (text)
- `membership\_type` (enum: SESSION_BASED, CREDIT_BASED, UNLIMITED, TIME_BASED)
- `sessions\_count` (integer, if SESSION_BASED)
- `credits\_count` (integer, if CREDIT_BASED)
- `validity\_days` (integer, e.g., 30)
- `price` (decimal, 8,2)
- `renewal\_reminder\_days` (integer, before expiry)
- `auto\_renew` (boolean, optional)
- `discount\_percentage` (decimal, 3,1, vs. per-session rate)
- `is\_active` (boolean)
- `created\_at` (timestamp)

CustomerMembership Model

****Fields:****

- `id` (UUID)
- `customer\_user` (FK to User)
- `membership` (FK to Membership)
- `start\_date` (date)
- `end\_date` (date)
- `sessions\_used` (integer, denormalized)
- `credits\_used` (integer, denormalized)
- `is\_active` (boolean)
- `status` (enum: ACTIVE, EXPIRED, CANCELLED, FROZEN)
- `renewal\_last\_notified` (timestamp)

1.9 Audit & Logging Models

AuditLog Model

****Fields:****

- `id` (UUID)
- `user` (FK to User, who performed action)
- `action` (enum: CREATE, UPDATE, DELETE, LOGIN, LOGOUT, OFFER_CREATED, COMMISSION_UPDATED, ...)
- `entity\_type` (string, e.g., 'Vendor', 'Booking', 'Offer')
- `entity\_id` (UUID)
- `old\_values` (JSON, before change)
- `new\_values` (JSON, after change)
- `ip\_address` (string)
- `user\_agent` (string)
- `timestamp` (timestamp, auto)
- `status` (enum: SUCCESS, FAILURE)
- `error\_message` (text, nullable)

LoginAttempt Model

****Fields:****

- `id` (UUID)
- `user` (FK to User)
- `ip\_address` (string)
- `user\_agent` (string)
- `success` (boolean)
- `reason\_failed` (string, nullable)
- `timestamp` (timestamp)

PART 2: API SPECIFICATION (Django REST Framework)

2.1 Authentication APIs

POST /api/v1/auth/register/

****Request:****

```json

```
{
 "email": "vendor@example.com",
 "phone_number": "+919876543210",
 "password": "SecurePass@123",
 "first_name": "Raj",
 "last_name": "Sharma",
 "role": "vendor"
}
```

...

**\*\*Response (201):\*\***

```json

```
{
  "id": "uuid-xxx",
  "email": "vendor@example.com",
  "role": "vendor",
  "is_verified": false,
  "message": "OTP sent to registered phone/email"
}
```

...

****Error (400):****

```json

```
{
 "error": "Email already registered",
 "field": "email"
}
```

...

#### #### POST /api/v1/auth/verify-otp/

**\*\*Request:\*\***

```json

```
{
  "email": "vendor@example.com",
  "otp": "123456"
}
```

```

...
**Response (200):**
```json
{
 "access_token": "jwt_token_xxx",
 "refresh_token": "refresh_token_xxx",
 "expires_in": 3600,
 "user": { "id": "uuid", "email": "...", "role": "vendor" }
}
...

```

```

POST /api/v1/auth/login/
Request:
```json
{
  "email": "vendor@example.com",
  "password": "SecurePass@123",
  "2fa_code": "123456" (optional, if 2FA enabled)
}
...

```

```

**Response (200):**
```json
{
 "access_token": "jwt_token_xxx",
 "refresh_token": "refresh_token_xxx",
 "user": { "id": "uuid", "email": "...", "role": "vendor" }
}
...

```

```

POST /api/v1/auth/refresh/
Request:
```json
{
  "refresh_token": "refresh_token_xxx"
}
...

```

```

**Response (200):**
```json
{
 "access_token": "new_jwt_token_xxx",
 "expires_in": 3600
}
...

```

```

POST /api/v1/auth/logout/
Response (200):
```json
{
  "message": "Logged out successfully",
  "session_ended_at": "timestamp"
}
...

```

```

#### POST /api/v1/auth/setup-2fa/

```

```
**Response (200):**
```json
{
 "qr_code": "data:image/png;base64,...",
 "secret": "JBSWY3DPEBLW64TMMQ=====",
 "backup_codes": ["code1", "code2", ...]
}
```
```

2.2 Super Admin APIs (Role-Based Access)

GET /api/v1/super-admin/admins/

Headers: `Authorization: Bearer jwt\_token`, `X-User-Role: super\_admin`

```
**Response (200):**
```json
{
 "count": 5,
 "results": [
 {
 "id": "uuid",
 "user": { "id": "uuid", "email": "admin@example.com",
"first_name": "Admin" },
 "vendor_access": ["all"], // or array of vendor IDs
 "permissions": { "can_onboard_vendor": true,
"can_manage_customers": true, ... },
 "created_at": "2025-01-01T10:00:00Z",
 "assigned_at": "2025-01-01T10:00:00Z"
 }
]
}
```
```

POST /api/v1/super-admin/admins/

```
**Request:**
```json
{
 "email": "newadmin@example.com",
 "first_name": "John",
 "last_name": "Doe",
 "phone_number": "+919876543210",
 "vendor_access": ["all"], // or ["vendor_id_1", "vendor_id_2"]
 "permissions": {
 "can_onboard_vendor": true,
 "can_manage_customers": true,
 "can_view_analytics": true,
 "can_moderate_offers": true
 }
}
```
```

```
**Response (201):**
```json
```

```
{
 "id": "uuid",
 "user": { "id": "uuid", "email": "newadmin@example.com" },
 "temp_password_sent": true,
 "message": "Admin created. Temporary password sent via email."
}
...
```

```
GET /api/v1/super-admin/offers/
Query Params: `?active=true&category=VENUE&page=1&limit=20`
Response (200):
```json
{
  "count": 50,
  "results": [
    {
      "id": "uuid",
      "name": "Welcome 20% Off",
      "offer_type": "INITIAL",
      "discount_type": "PERCENTAGE",
      "discount_value": 20,
      "visibility_percentage": 30,
      "applicable_categories": ["VENUE", "COACH"],
      "is_active": true,
      "start_date": "2025-01-01",
      "end_date": "2025-12-31",
      "redemption_count": 1250,
      "revenue_impact": 50000.00,
      "created_by": { "email": "super_admin@example.com" }
    }
  ]
}
...
```

```
#### POST /api/v1/super-admin/offers/
**Request:**
```json
{
 "name": "Festive Diwali Offer",
 "offer_type": "FESTIVE",
 "discount_type": "PERCENTAGE",
 "discount_value": 25,
 "max_discount_cap": 5000,
 "minimum_booking_amount": 1000,
 "applicable_categories": ["VENUE", "COACH", "ECOMMERCE"],
 "start_date": "2025-11-01",
 "end_date": "2025-11-15",
 "visibility_percentage": 20,
 "usage_limit_total": 5000,
 "usage_limit_per_customer": 3
}
...

Response (201):
```json
```

```
{
  "id": "uuid",
  "name": "Festive Diwali Offer",
  "is_active": true,
  "created_at": "2025-01-01T10:00:00Z"
}
...
```

PUT /api/v1/super-admin/offers/{offer_id}/

Request:

```json

```
{
 "is_active": false
}
...
```

\*\*Response (200):\*\*

```json

```
{
  "id": "uuid",
  "is_active": false,
  "updated_at": "2025-01-01T11:00:00Z"
}
...
```

GET /api/v1/super-admin/commission-config/

Response (200):

```json

```
{
 "id": "uuid",
 "default_commission_percentage": 20.00,
 "category_overrides": {
 "VENUE": 18.0,
 "COACH": 22.0,
 "ECOMMERCE": 20.0
 },
 "vendor_custom_rates": {
 "vendor_id_1": 15.0,
 "vendor_id_2": 18.0
 },
 "payout_cycle": "WEEKLY",
 "minimum_payout_threshold": 5000.00,
 "tax_rate_gst": 18.0
}
...
```

#### PUT /api/v1/super-admin/commission-config/

\*\*Request:\*\*

```json

```
{
  "default_commission_percentage": 22.0,
  "category_overrides": {
    "VENUE": 20.0,
    "COACH": 25.0,
    "ECOMMERCE": 20.0
  }
}
```

```
    },  
    "payout_cycle": "WEEKLY"  
  }  
  ...
```

```
**Response (200):**  
```json  
{
 "id": "uuid",
 "default_commission_percentage": 22.0,
 "updated_at": "2025-01-01T11:00:00Z"
}
...
```

```
GET /api/v1/super-admin/analytics/
Query Params: `?
start_date=2025-01-01&end_date=2025-01-31&metric=revenue`
Response (200):
```json  
{  
  "total_revenue": 5000000.00,  
  "total_commission": 1000000.00,  
  "total_payouts": 4000000.00,  
  "total_bookings": 15000,  
  "total_users": 50000,  
  "daily_breakdown": [  
    {  
      "date": "2025-01-01",  
      "revenue": 150000.00,  
      "commission": 30000.00,  
      "bookings": 450  
    }  
  ],  
  "category_breakdown": {  
    "VENUE": { "revenue": 3000000.00, "commission": 600000.00 },  
    "COACH": { "revenue": 1500000.00, "commission": 330000.00 },  
    "ECOMMERCE": { "revenue": 500000.00, "commission": 100000.00 }  
  }  
}  
...
```

```
#### GET /api/v1/super-admin/audit-logs/  
**Query Params:** `?  
user_id=uuid&action=CREATE&entity_type=Vendor&page=1&limit=50`  
**Response (200):**  
```json  
{
 "count": 1000,
 "results": [
 {
 "id": "uuid",
 "user": { "email": "admin@example.com" },
 "action": "CREATE",
 "entity_type": "Vendor",
 "entity_id": "vendor_uuid",
 }
]
}
```



```

 "old_values": null,
 "new_values": { "vendor_name": "New Venue", "city":
"Bangalore" },
 "ip_address": "192.168.1.1",
 "timestamp": "2025-01-01T10:00:00Z",
 "status": "SUCCESS"
 }
]
}
...

```

---

### ### 2.3 Admin APIs (Role-Based Access)

```

GET /api/v1/admin/vendors/
Headers: `Authorization: Bearer jwt_token`, `X-User-Role: admin`
Query Params: `?
vendor_type=VENUE&approval_status=PENDING&page=1&limit=20`
Response (200):
```json
{
  "count": 150,
  "results": [
    {
      "id": "vendor_uuid",
      "business_name": "Court Central",
      "vendor_type": "VENUE",
      "approval_status": "PENDING",
      "kyc_status": "VERIFIED",
      "legal_name": "Court Central Pvt Ltd",
      "city": "Bangalore",
      "onboarded_at": "2025-01-15T10:00:00Z",
      "onboarded_by": { "email": "admin@example.com" },
      "total_bookings": 500,
      "total_revenue": 250000.00,
      "avg_rating": 4.8,
      "commission_rate": 20.0
    }
  ]
}
...

```

```

#### PUT /api/v1/admin/vendors/{vendor_id}/approve/
**Request:**
```json
{
 "approval_status": "APPROVED",
 "notes": "KYC verified, documentation complete"
}
...
Response (200):
```json
{

```

```
    "id": "vendor_uuid",
    "approval_status": "APPROVED",
    "approved_at": "2025-01-15T11:00:00Z",
    "approved_by": { "email": "admin@example.com" }
  },
  ...
```

PUT /api/v1/admin/vendors/{vendor_id}/reject/

Request:

```json

```
{
 "approval_status": "REJECTED",
 "rejection_reason": "KYC documents incomplete"
},
...
```

\*\*Response (200):\*\*

```json

```
{
  "id": "vendor_uuid",
  "approval_status": "REJECTED",
  "rejection_reason": "KYC documents incomplete"
},
...
```

GET /api/v1/admin/customers/

Query Params: `?search=john&kyc\_status=VERIFIED&page=1&limit=20`

Response (200):

```json

```
{
 "count": 5000,
 "results": [
 {
 "id": "customer_uuid",
 "user": { "email": "customer@example.com", "first_name":
"John", "phone_number": "+919876543210" },
 "total_bookings": 15,
 "total_spending": 45000.00,
 "first_booking_date": "2025-01-01",
 "last_booking_date": "2025-01-20",
 "kyc_verified": true
 }
]
},
...
```

#### POST /api/v1/admin/vendors/onboard/

\*\*Request:\*\*

```json

```
{
  "business_name": "New Court Venue",
  "vendor_type": "VENUE",
  "legal_name": "New Court Venue Pvt Ltd",
  "email": "newcourt@example.com",
  "phone_number": "+919876543210",
}
```

```
    "city": "Delhi",
    "state": "Delhi",
    "postal_code": "110001",
    "pan_number": "ABCDE1234F",
    "gst_in": "07AABCU9603R1Z0"
  }
}
```

Response (201):

```json

```
{
 "id": "vendor_uuid",
 "business_name": "New Court Venue",
 "approval_status": "PENDING",
 "onboarded_by": { "email": "admin@example.com" },
 "message": "Vendor onboarded successfully. Awaiting KYC
verification."
}
```

---

### ### 2.4 Vendor APIs (Mobile + Web)

#### GET /api/v1/vendor/dashboard/

\*\*\*Headers:\*\* `Authorization: Bearer jwt\_token`, `X-User-Role: vendor`, `X-Vendor-ID: vendor\_uuid`

\*\*\*Response (200):\*\*\*

```json

```
{
  "today_bookings": 12,
  "today_earnings": 45000.00,
  "pending_payout": 50000.00,
  "total_customers": 150,
  "active_courts": 5,
  "avg_rating": 4.8,
  "occupancy_percentage": 78,
  "this_month_revenue": 1200000.00,
  "active_offers_count": 3,
  "active_memberships_count": 25,
  "quick_actions": [
    { "action": "add_court", "icon": "plus", "label": "Add Court" },
    { "action": "view_bookings", "icon": "calendar", "label": "View
Bookings" },
    { "action": "manage_offers", "icon": "tag", "label": "Manage
Offers" }
  ]
}
```

POST /api/v1/vendor/venues/

***Request:**

```json

```
{
 "name": "Central Sports Complex",
```

```
 "description": "Professional sports facility in heart of city",
 "address": "123 Main Street, Bangalore",
 "city": "Bangalore",
 "state": "Karnataka",
 "postal_code": "560001",
 "latitude": 13.0827,
 "longitude": 80.2707,
 "phone_number": "+919876543210",
 "opening_time": "06:00",
 "closing_time": "22:00",
 "is_open_24_7": false,
 "amenities": ["WiFi", "Parking", "Cafeteria", "Changing Rooms"],
 "cancellation_window_hours": 24,
 "reschedule_window_hours": 48,
 "holiday_dates": ["2025-12-25", "2025-01-26"],
 "images": ["url1", "url2", "url3"]
 }
 ...
```

```
Response (201):
```json
{
  "id": "venue_uuid",
  "name": "Central Sports Complex",
  "is_verified": false,
  "created_at": "2025-01-15T10:00:00Z"
}
...
```

POST /api/v1/vendor/courts/

```
**Request:**
```json
{
 "venue_id": "venue_uuid",
 "name": "Cricket Court A",
 "court_type": "CRICKET",
 "base_price_per_hour": 1500.00,
 "capacity": 22,
 "lights_available": true,
 "equipment_available": ["Stumps", "Bats", "Balls"],
 "surface_type": "Pitch",
 "images": ["url1", "url2"]
}
...
```

```
Response (201):
```json
{
  "id": "court_uuid",
  "name": "Cricket Court A",
  "venue_id": "venue_uuid",
  "base_price_per_hour": 1500.00,
  "created_at": "2025-01-15T10:00:00Z"
}
...
```

```
#### POST /api/v1/vendor/courts/{court_id}/availability/
**Request:**
```json
{
 "availabilities": [
 { "day_of_week": 0, "start_time": "06:00", "end_time": "22:00",
 "is_available": true },
 { "day_of_week": 1, "start_time": "06:00", "end_time": "22:00",
 "is_available": true }
]
}
```
**Response (200):**
```json
{
 "id": "court_uuid",
 "availabilities_updated": 7,
 "message": "Court availability updated"
}
```
```

```
#### POST /api/v1/vendor/courts/{court_id}/slots/generate/
**Request:**
```json
{
 "date": "2025-01-20",
 "slot_duration_minutes": 60,
 "start_time": "06:00",
 "end_time": "22:00"
}
```
**Response (200):**
```json
{
 "date": "2025-01-20",
 "slots_generated": 16,
 "court_id": "court_uuid"
}
```
```

```
#### PUT /api/v1/vendor/courts/{court_id}/pricing/
**Request:**
```json
{
 "base_price_per_hour": 1500.00,
 "time_slots": [
 { "start_time": "06:00", "end_time": "12:00", "price": 1200.00,
 "type": "MORNING" },
 { "start_time": "12:00", "end_time": "18:00", "price": 1500.00,
 "type": "AFTERNOON" },
 { "start_time": "18:00", "end_time": "22:00", "price": 1800.00,
 "type": "EVENING" }
]
}
```
```

```

\*\*Response (200):\*\*

```json

```
{
  "id": "court_uuid",
  "pricing_updated": true,
  "effective_from": "2025-01-16T00:00:00Z"
}
```

```

#### GET /api/v1/vendor/bookings/

\*\*Query Params:\*\* `?

date=2025-01-20&status=CONFIRMED&court\_id=court\_uuid&page=1&limit=20

,

\*\*Response (200):\*\*

```json

```
{
  "count": 150,
  "today_count": 12,
  "results": [
    {
      "id": "booking_uuid",
      "court": { "id": "court_uuid", "name": "Cricket Court A" },
      "customer_name": "John Doe",
      "customer_phone": "+919876543210",
      "date": "2025-01-20",
      "start_time": "10:00",
      "end_time": "11:00",
      "base_price": 1500.00,
      "applied_offer": { "id": "offer_uuid", "name": "Welcome 20%
Off", "discount_amount": 300.00 },
      "total_amount": 1416.00,
      "commission_amount": 283.20,
      "vendor_payout": 1132.80,
      "payment_status": "PAID",
      "booking_status": "CONFIRMED",
      "internal_notes": "Customer requested evening slot, shifted to
morning",
      "customer_notes": "Bring extra balls",
      "created_at": "2025-01-15T10:00:00Z"
    }
  ]
}
```

```

#### GET /api/v1/vendor/bookings/{booking\_id}/

\*\*Response (200):\*\*

```json

```
{
  "id": "booking_uuid",
  "court": { "id": "court_uuid", "name": "Cricket Court A", "venue":
{ "id": "venue_uuid", "name": "Central Sports Complex" } },
  "customer": {
    "id": "customer_uuid",
```

```

        "name": "John Doe",
        "phone": "+919876543210",
        "email": "john@example.com",
        "total_bookings": 15,
        "avg_rating": 4.8,
        "member_since": "2024-06-01"
    },
    "date": "2025-01-20",
    "start_time": "10:00",
    "end_time": "11:00",
    "duration_minutes": 60,
    "base_price": 1500.00,
    "discount_amount": 300.00,
    "applied_offer": { "id": "offer_uuid", "name": "Welcome 20% Off",
"offer_type": "INITIAL" },
    "tax_percentage": 18.0,
    "tax_amount": 216.00,
    "commission_percentage": 20.0,
    "commission_amount": 283.20,
    "total_amount": 1416.00,
    "vendor_payout": 1132.80,
    "payment_method": "ONLINE",
    "payment_status": "PAID",
    "payment_id": "razorpay_payment_id",
    "booking_status": "CONFIRMED",
    "customer_notes": "Bring extra balls",
    "internal_notes": "Customer requested evening slot, shifted to
morning",
    "can_reschedule": true,
    "can_cancel": true,
    "rescheduling_window_hours": 48,
    "cancellation_window_hours": 24
}
...

```

PUT /api/v1/vendor/bookings/{booking_id}/reschedule/

Request:

```json

```

{
 "new_date": "2025-01-21",
 "new_start_time": "14:00",
 "new_end_time": "15:00"
}
...

```

\*\*Response (200):\*\*

```json

```

{
  "id": "booking_uuid",
  "rescheduled": true,
  "new_date": "2025-01-21",
  "new_start_time": "14:00",
  "rescheduled_at": "2025-01-15T11:00:00Z"
}
...

```

POST /api/v1/vendor/bookings/{booking_id}/cancel/

Request:

```json

```
{
 "reason": "Customer requested cancellation"
}
```

```

Response (200):

```json

```
{
 "id": "booking_uuid",
 "status": "CANCELLED",
 "refund_amount": 1416.00,
 "refund_status": "PROCESSED",
 "cancelled_at": "2025-01-15T11:00:00Z"
}
```

```

GET /api/v1/vendor/customers/

Query Params: `?

search=john&sort_by=last_booking&page=1&limit=20`

Response (200):

```json

```
{
 "count": 150,
 "results": [
 {
 "id": "customer_uuid",
 "name": "John Doe",
 "phone": "+919876543210",
 "email": "john@example.com",
 "total_bookings": 15,
 "total_spent": 45000.00,
 "first_booking": "2024-06-01",
 "last_booking": "2025-01-20",
 "avg_rating": 4.8,
 "membership_status": { "id": "uuid", "name": "Monthly Pass",
"expires_at": "2025-02-15" },
 "tags": ["Regular", "VIP"]
 }
]
}
```

```

POST /api/v1/vendor/memberships/

Request:

```json

```
{
 "name": "Monthly Court Pass",
 "description": "Unlimited access to all courts for 30 days",
 "membership_type": "UNLIMITED",
 "validity_days": 30,
 "price": 10000.00,
}
```



```
 "discount_percentage": 15,
 "auto_renew": true,
 "renewal_reminder_days": 3
 },
 ...
Response (201):
```json
{
  "id": "membership_uuid",
  "name": "Monthly Court Pass",
  "price": 10000.00,
  "created_at": "2025-01-15T10:00:00Z"
}
...
```

```
#### GET /api/v1/vendor/offers/
**Query Params:** `?is_active=true&page=1&limit=20`
**Response (200):**
```json
{
 "count": 5,
 "results": [
 {
 "id": "offer_uuid",
 "name": "Welcome 20% Off",
 "offer_type": "INITIAL",
 "discount_type": "PERCENTAGE",
 "discount_value": 20,
 "is_active": true,
 "start_date": "2025-01-01",
 "end_date": "2025-12-31",
 "usage_limit_total": 5000,
 "usage_limit_per_customer": 3,
 "redemption_count": 1250,
 "created_by": { "email": "super_admin@example.com" }
 }
]
}
...
```

```
GET /api/v1/vendor/financial/summary/
Query Params: `?
start_date=2025-01-01&end_date=2025-01-31&period=DAILY`
Response (200):
```json
{
  "period": "JANUARY 2025",
  "total_gross_revenue": 1200000.00,
  "total_commission": 240000.00,
  "total_refunds": 15000.00,
  "net_revenue": 945000.00,
  "total_payouts": 945000.00,
  "pending_payout": 50000.00,
  "next_payout_date": "2025-02-07",
}
```

```

    "breakdown": [
      {
        "date": "2025-01-01",
        "bookings": 45,
        "revenue": 40000.00,
        "commission": 8000.00,
        "payout": 32000.00
      }
    ]
  }
}
...

```

```

#### GET /api/v1/vendor/analytics/
**Query Params:** `?
metric=bookings&start_date=2025-01-01&end_date=2025-01-31&groupby=daily`
**Response (200):**
```json
{
 "metric": "bookings",
 "total": 500,
 "daily_average": 16.7,
 "trend": "UP",
 "data": [
 { "date": "2025-01-01", "value": 15, "revenue": 40000.00 },
 { "date": "2025-01-02", "value": 18, "revenue": 45000.00 }
],
 "top_services": [
 { "name": "Cricket Court A", "bookings": 150, "revenue": 225000.00 },
 { "name": "Badminton Court B", "bookings": 120, "revenue": 180000.00 }
]
}
...

```

---

### ### 2.5 Coach APIs

```

POST /api/v1/vendor/coaches/
Request:
```json
{
  "first_name": "Rajesh",
  "last_name": "Sharma",
  "bio": "15 years of cricket coaching experience",
  "specializations": ["Cricket - Batting", "Cricket - Bowling"],
  "years_of_experience": 15,
  "certifications": [
    { "name": "Level 2 Cricket Coach", "issuer": "ICC", "number": "ICC12345", "doc_url": "url" }
  ],
  "hourly_rate": 1000.00,

```

```
    "available_for_online": true,
    "available_for_venue": true,
    "available_for_home": true,
    "max_home_travel_km": 15,
    "home_visit_extra_charge": 200.00
  }
  ...
```

****Response (201):****

```
```json
{
 "id": "coach_uuid",
 "first_name": "Rajesh",
 "verification_status": "PENDING",
 "message": "Coach profile created. Awaiting admin verification."
}
...

```

**#### POST /api/v1/vendor/coaches/{coach\_id}/availability/**

**\*\*Request:\*\***

```
```json
{
  "availabilities": [
    { "day_of_week": 0, "start_time": "07:00", "end_time": "19:00",
    "recurring": true },
    { "day_of_week": 6, "start_time": "08:00", "end_time": "20:00",
    "recurring": true }
  ],
  "lead_time_hours": 6
}
...

```

****Response (200):****

```
```json
{
 "id": "coach_uuid",
 "availabilities_updated": 7
}
...

```

**#### GET /api/v1/vendor/coaching-sessions/**

**\*\*Query Params:\*\*** `?

date=2025-01-20&status=CONFIRMED&page=1&limit=20`

**\*\*Response (200):\*\***

```
```json
{
  "count": 50,
  "results": [
    {
      "id": "session_uuid",
      "coach": { "id": "coach_uuid", "name": "Rajesh Sharma" },
      "customer": { "id": "customer_uuid", "name": "John Doe",
      "phone": "+919876543210" },
      "session_type": "1-on-1 Cricket Batting",
      "date": "2025-01-20",
      "start_time": "10:00",

```

```

        "end_time": "11:00",
        "delivery_mode": "VENUE",
        "base_price": 1000.00,
        "applied_offer": { "discount_amount": 200.00 },
        "total_amount": 945.00,
        "commission_amount": 189.00,
        "vendor_payout": 756.00,
        "payment_status": "PAID",
        "attendance_status": "COMPLETED",
        "feedback_rating": 4.8
    }
]
}
\..

```

2.6 Ecommerce APIs

POST /api/v1/vendor/products/

****Request:****

```json

```

{
 "sku": "CRICKET_BAT_001",
 "name": "Premium Cricket Bat – English Willow",
 "description": "Professional grade English willow cricket bat",
 "category": "EQUIPMENT",
 "price": 15000.00,
 "cost_price": 8000.00,
 "discount_percentage": 10.0,
 "currency": "INR",
 "stock_quantity": 50,
 "low_stock_threshold": 10,
 "images": ["url1", "url2", "url3"],
 "weight_kg": 1.25,
 "dimensions_cm": { "length": 96, "width": 4.5, "height": 3 },
 "shipping_class": "REGULAR",
 "is_returnable": true,
 "return_window_days": 30,
 "warranty_description": "1 year manufacturing defect warranty"
}
\..

```

**\*\*Response (201):\*\***

```json

```

{
  "id": "product_uuid",
  "sku": "CRICKET_BAT_001",
  "name": "Premium Cricket Bat – English Willow",
  "created_at": "2025-01-15T10:00:00Z"
}
\..

```

GET /api/v1/vendor/products/

****Query Params:**** `?

category=EQUIPMENT&is_active=true&stock_status=in_stock&page=1&limit=20`

****Response (200):****

```json

```
{
 "count": 150,
 "results": [
 {
 "id": "product_uuid",
 "sku": "CRICKET_BAT_001",
 "name": "Premium Cricket Bat",
 "price": 15000.00,
 "stock_quantity": 50,
 "low_stock": false,
 "sales_count": 120,
 "revenue": 1800000.00,
 "avg_rating": 4.7,
 "is_active": true
 }
]
}
```

```

GET /api/v1/vendor/orders/

****Query Params:**** `?

status=SHIPPED&payment_status=PAID&page=1&limit=20`

****Response (200):****

```json

```
{
 "count": 500,
 "results": [
 {
 "id": "order_uuid",
 "order_number": "ORD-20250115-001",
 "customer": { "id": "customer_uuid", "name": "John Doe",
"phone": "+919876543210" },
 "items": [
 { "product": { "id": "product_uuid", "name": "Cricket Bat",
"sku": "CRICKET_BAT_001" }, "quantity": 1, "price": 15000.00 }
],
 "subtotal": 15000.00,
 "discount_amount": 1500.00,
 "tax_amount": 2430.00,
 "total_amount": 15930.00,
 "commission_amount": 3186.00,
 "vendor_payout": 12744.00,
 "payment_status": "PAID",
 "order_status": "SHIPPED",
 "tracking_number": "TRK123456789",
 "courier_name": "Shiprocket",
 "shipping_address": { "name": "John Doe", "city": "Bangalore",
"state": "Karnataka" },
 "created_at": "2025-01-15T10:00:00Z"
 }
]
}
```

```
]
}
...
```

---

## ## PART 3: TECHNICAL FLOWS (Data, Payment, Auth)

### ### 3.1 Authentication Flow

...

USER (Web/Mobile)

```
↓
POST /api/v1/auth/register/
↓
Backend: Create User, send OTP
↓
POST /api/v1/auth/verify-otp/
↓
Backend: Verify OTP, generate JWT + Refresh Token
↓
User: Store JWT in secure storage (HttpOnly cookie for web,
Keychain/Keystore for mobile)
↓
All subsequent requests: Include JWT in Authorization header
↓
If JWT expires → POST /api/v1/auth/refresh/ with Refresh Token
↓
Backend: Validate Refresh Token, issue new JWT
↓
Continue API calls
...
```

### ### 3.2 Booking Flow

...

VENDOR (Mobile App)

```
↓
Vendor dashboard → Tap "New Booking"
↓
POST /api/v1/vendor/bookings/create/
{
 "court_id": "court_uuid",
 "customer_phone": "+919876543210",
 "date": "2025-01-20",
 "start_time": "10:00",
 "end_time": "11:00"
}
↓
Backend:
 1. Fetch Court, TimeSlot, Customer
 2. Calculate base_price
 3. Fetch applicable offers (INITIAL if new customer, FESTIVE
if date matches)
```

```

 4. Apply best offer (top N%)
 5. Calculate tax (18% GST)
 6. Calculate commission (20% default, or admin override)
 7. Calculate vendor_payout
 8. Create Booking record
 9. Return booking_id
↓
Response:
{
 "id": "booking_uuid",
 "base_price": 1500.00,
 "discount_amount": 300.00,
 "applied_offer": { "id": "offer_uuid", "name": "Welcome 20%
Off" },
 "total_amount": 1416.00,
 "requires_payment": true
}
↓
Vendor selects payment method:
- If CASH: Mark booking CONFIRMED, no payment call
- If ONLINE: Call POST /api/v1/vendor/bookings/{booking_id}/
pay/
↓
Backend (if ONLINE):
 1. Initialize Razorpay checkout
 2. Return payment_order_id
↓
Frontend: Show Razorpay checkout modal
↓
Customer pays (or vendor enters for offline booking)
↓
Razorpay → Webhook to Backend:
 POST /api/v1/webhooks/razorpay/payment_success/
 {
 "payment_id": "pay_xxx",
 "order_id": "order_xxx",
 "booking_id": "booking_uuid"
 }
↓
Backend:
 1. Verify payment signature
 2. Update Booking: payment_status = PAID, booking_status =
CONFIRMED
 3. Send SMS/Email confirmation to customer
 4. Send notification to vendor
 5. Emit event (booking.confirmed) for analytics
↓
Vendor sees booking in Dashboard: status = CONFIRMED, payment =
PAID
```

```

3.3 Offer Application Flow

```
```
```

SUPER ADMIN (Web)

↓

Create Offer:

POST /api/v1/super-admin/offers/

```
{
 "name": "Welcome 20% Off",
 "offer_type": "INITIAL",
 "discount_type": "PERCENTAGE",
 "discount_value": 20,
 "applicable_categories": ["VENUE", "COACH", "ECOMMERCE"],
 "start_date": "2025-01-01",
 "end_date": "2025-12-31",
 "visibility_percentage": 30,
 "usage_limit_per_customer": 3
}
```

↓

Backend:

1. Create Offer record
2. Mark is\_active = true
3. Set visibility\_percentage = 30 (show in top 30% of offers)

↓

When CUSTOMER BOOKS:

↓

Backend fetches applicable offers:

1. Get all active offers for category (VENUE/COACH/ECOMMERCE)
2. Check offer validity (date, time, day\_of\_week)
3. Check customer eligibility (new, not in exclude list, usage limit not exceeded)
4. Filter by visibility\_percentage (e.g., if 30%, show only best offers that top 30%)
5. Sort by discount\_value (highest first)
6. Apply best offer automatically OR show top 3 offers for customer to choose

↓

Calculate:

discount\_amount = base\_price \* (discount\_value / 100) capped at max\_discount\_cap

taxable\_amount = base\_price - discount\_amount

tax\_amount = taxable\_amount \* (tax\_percentage / 100)

total\_amount = taxable\_amount + tax\_amount

↓

Create Booking with applied\_offer\_id

↓

Log redemption: Offer.redemption\_count += 1

...

### ### 3.4 Commission & Payout Flow

...

BOOKING/ORDER CREATED:

↓

Backend calculates commission:

1. Fetch CommissionConfig



```

 2. Get default_commission_percentage (20%)
 3. Check category_overrides (VENUE=18%, COACH=22%, etc.)
 4. Check vendor_custom_rates (if specific vendor has custom
rate)
 5. commission_percentage = determined_rate
 6. commission_amount = total_amount * (commission_percentage /
100)
 7. vendor_payout = total_amount - commission_amount
↓
Log transaction:
{
 "booking_id": "booking_uuid",
 "vendor_id": "vendor_uuid",
 "gross_amount": total_amount,
 "commission_amount": commission_amount,
 "vendor_payout": vendor_payout,
 "status": "PENDING"
}
↓
Every day/week (per payout_cycle):
 1. Aggregate pending transactions for vendor
 2. If payout_amount >= minimum_payout_threshold:
 a. Create Payout record
 b. Transfer to vendor's bank account (Razorpay/payment
provider)
 c. Mark transactions status = PROCESSED
 d. Send email: "Payout processed: ₹XXXX"
↓
Vendor sees in Dashboard:
 today_payout: 0
 pending_payout: 50000
 last_payout: 100000 (processed on 2025-01-14)
 next_payout_date: 2025-01-21
...

```

### ### 3.5 Customer Eligibility for Offers

```

...
When displaying offers to CUSTOMER:
↓
Check offer applicability:
 1. Is offer ACTIVE? (is_active = true)
 2. Is today within start_date and end_date?
 3. Is current_time within start_time and end_time (if
specified)?
 4. Is today in applicable_days_of_week?
 5. Is booking_amount >= minimum_booking_amount?
 6. Is customer in include_customer_ids (if list specified)?
 7. Is customer NOT in exclude_customer_ids?
 8. Has customer used this offer < usage_limit_per_customer?
 9. Has offer been redeemed < usage_limit_total times?
↓
If ALL checks pass: Offer is applicable
↓

```

```

 Filter offers by visibility_percentage:
 1. If visibility_percentage = 30, show top 30% by
discount_value
 2. Sort all applicable offers by discount value (highest
first)
 3. Take top 30% from sorted list
 ↓
 Show to customer (auto-apply best or let customer choose)
...

```

---

## ## PART 4: MOBILE APP SCREENS & NAVIGATION (React Native/Expo)

### ### 4.1 Authentication Screens

#### #### LoginScreen

**\*\*Route:\*\*** `/auth/login`

**\*\*Components:\*\***

- Header: Logo + "Sign In"
- Tabs: Email/Phone toggle
- TextInput: Email or Phone Number
- TextInput: Password (secure)
- Button: "Sign In"
- Checkbox: "Remember me"
- Link: "Forgot password?"
- Link: "Sign up" → RegisterScreen
- Divider
- Button: "Continue with Google" (Google Sign-In)
- Button: "Continue with Apple" (Apple Sign-In)
- Error message display (red text)
- Loading indicator (while calling API)

**\*\*State:\*\***

```javascript

```

{
  loginMethod: "email", // or "phone"
  email: "",
  phoneNumber: "",
  password: "",
  rememberMe: false,
  loading: false,
  error: ""
}
...

```

****API Call:****

...

POST /api/v1/auth/login/

```

{
  "email": "vendor@example.com",
  "password": "SecurePass@123"
}

```

→ Response: { access_token, refresh_token, user }

→ Store in secure storage, navigate to Dashboard
```

#### #### RegisterScreen

**\*\*Route:\*\*** `/auth/register`

**\*\*Screens (Stepper):\*\***

**\*\*Step 1: Account Type\*\***

- Radio buttons: Vendor / Customer
- Description per type
- Next button

**\*\*Step 2: Basic Info\*\***

- TextInput: First Name
- TextInput: Last Name
- TextInput: Email
- TextInput: Phone Number
- TextInput: Password (validation: 12 chars, 1 upper, 1 lower, 1 digit, 1 special)
- TextInput: Confirm Password
- Checkbox: Agree to Terms
- Button: Sign Up

**\*\*Step 3: OTP Verification\*\***

- TextInput: Enter OTP (6 digits)
- Button: Verify OTP
- Button: Resend OTP (with 30-second countdown)
- Error/Success messages

**\*\*Step 4: Success\*\***

- Checkmark animation
- Message: "Account created successfully"
- Button: Go to Dashboard

#### #### SplashScreen

**\*\*Route:\*\*** `/splash` (shown on app startup)

**\*\*Components:\*\***

- Full screen with logo
- Loading animation (spinner or pulsing icon)
- Optional tagline: "Loading your dashboard..."

**\*\*Logic:\*\***

```javascript

```
useEffect(() => {
  setTimeout(() => {
    // Check if token exists in secure storage
    const token = secureStorage.getItem('access_token');
    if (token) {
      // User logged in → Dashboard
      navigation.replace('VendorDashboard');
    } else {
      // User not logged in → Login
      navigation.replace('LoginScreen');
    }
  })
})
```

```
    }, 2000); // 2-second splash duration
  }, []);
}, \
```

4.2 Vendor Dashboard Navigation

****Bottom Tab Navigation:****

\ \

| |
|--|
| Dashboard Calendar Bookings More |
|--|

\ \

DashboardScreen

****Route:**** `/vendor/dashboard`

****Components:****

****Header:****

- Profile icon (tap to show menu: Profile, Settings, Logout)
- Notification bell (with unread count badge)

****Scrollable Content:****

1. ****Today's Performance Cards**** (horizontal scroll):

- Card 1: "Today's Bookings" (count, occupancy %)
- Card 2: "Today's Earnings" (amount, pending)
- Card 3: "Active Customers" (count)
- Card 4: "Avg Rating" (stars, count of reviews)

2. ****Quick Actions Row****:

- Button: "+ Add Court"
- Button: "View Calendar"
- Button: "Manage Offers"
- Button: "View Orders" (if ecommerce)

3. ****Recent Activity****:

- List of last 5 bookings/orders
- Customer name, time, amount, status badge
- Tap to view detail

4. ****Active Memberships****:

- Chip: "8 Active Memberships"
- Tap to see details

5. ****Offers Performance****:

- "3 Active Offers"
- Top offer: discount, redemptions count

6. ****Financial Summary****:

- This month revenue (amount)
- Pending payout (amount)
- Last payout (amount, date)

****Navigation Actions:****

- Tap card → Detail screen
- Tap "+" button → Create flow
- Tap booking → BookingDetailScreen

CalendarScreen

****Route:**** `/vendor/calendar`

****Components:****

****Header:****

- Venue selector (dropdown)
- Date picker (shows current date, tap to change)
- View toggle: Day / Week (buttons)

****Filters (collapsible):****

- Sport filter (checkbox multi-select)
- Court filter (checkbox multi-select)
- Status filter: All / Booked / Available / Blocked

****Day View Layout:****

- Left column: Time slots (06:00 – 22:00, 30-min intervals)
- Right section: Court columns (Cricket Court A, Badminton Court B, etc.)
- Tap on time slot → see details or create booking

****Slot Appearance:****

- Free slot: Light green, "AVAILABLE – ₹1500"
- Booked slot: Blue, customer initials, amount
- Blocked slot: Red, "MAINTENANCE" or "BLOCKED"
- Maintenance: Gray, "CLOSED"

****Floating Action Buttons:****

- FAB: "+ Booking" (primary action)
- Menu: ":" (more options)
 - Block Slot
 - Bulk Operations
 - Settings

****Tap Free Slot:****

\\\

→ Show modal:

1. Slot details (date, time, court)
2. Customer phone/email input (search or create new)
3. Price (base_price – discount from active offers)
4. Payment method (Online / Cash)
5. "Create Booking" button

\\\

****Tap Booked Slot:****

\\\

→ Show modal:

1. Booking summary (customer, amount, status)

```

2. Actions:
  - View Detail
  - Reschedule
  - Cancel
  - Add Note
...

---

#### BookingsListScreen
**Route:** `/vendor/bookings`
**Components:**

**Header:**
- Filters (dropdown): Date range, Status, Payment status
- Search (by customer name)
- Sort (by date, amount, status)
- Export button (CSV)

**List:**
- FlatList: Scroll to load more
- Each item:
  - Customer name (avatar + name)
  - Date/Time (small text)
  - Duration (small text)
  - Amount (bold, ₹)
  - Status badge (colors: Confirmed=green, Pending=yellow,
Cancelled=red)
  - Tap → BookingDetailScreen

**Empty State:**
- "No bookings found"
- "Add your first booking" button

#### BookingDetailScreen
**Route:** `/vendor/bookings/{booking_id}`
**Components:**

**Header:**
- Back button
- Booking ID
- Status badge

**Scrollable Content:**
1. **Customer Info Card:**
  - Name, phone, email (clickable for call/SMS)
  - Total bookings (count), avg rating (stars)
  - Member since (date)

2. **Booking Summary:**
  - Court name
  - Date & time
  - Duration (minutes)
  - Delivery mode (if coaching)

```

3. ****Pricing Breakdown:****
 - Base price: ₹1500
 - Discount: -₹300 (offer name)
 - Subtotal: ₹1200
 - Tax (GST): +₹216
 - ****Total: ₹1416****
 - Commission: -₹283 (20%)
 - Your payout: ₹1133
4. ****Payment Details:****
 - Payment method: Online / Cash
 - Payment status: Paid / Pending / Refunded
 - Transaction ID (if online)
5. ****Status & Actions:****
 - Current status: Confirmed / Completed / Cancelled
 - If Confirmed:
 - Button: "Reschedule"
 - Button: "Cancel"
 - Button: "Check In" (if within 15 mins)
 - If Completed:
 - Button: "Request Review"
 - If Cancelled:
 - Show refund details
6. ****Internal Notes:****
 - Notes from vendor (read-only or editable if not completed)
 - Button: "Add Note"
7. ****Customer Notes:****
 - Special requests (read-only)
8. ****Audit Info:****
 - Created: date/time
 - Last updated: date/time

4.3 Vendor Setup Screens

VenueRegistrationScreen

****Route:**** `/vendor/setup/venue`

****Flow (Stepper):****

****Step 1: Venue Basics****

- TextInput: Venue Name
- TextInput: Description (textarea)
- TextInput: Address
- TextInput: City
- TextInput: State (dropdown)
- TextInput: Postal Code
- TextInput: Phone Number
- Touchable: "Set Location on Map" (Google Maps)

- Latitude/Longitude (read-only, auto-filled)

****Step 2: Operating Hours****

- Time Picker: Opening Time (default 06:00)
- Time Picker: Closing Time (default 22:00)
- Toggle: "24/7 Open" (overrides time picker)
- Multi-date picker: Holiday dates
- TextInput: Cancellation window (hours)
- TextInput: Reschedule window (hours)

****Step 3: Amenities & Details****

- Checkbox group: Amenities (WiFi, Parking, Cafeteria, Changing Rooms, Washrooms, AC, Lights, Equipment Rental, First Aid)
- Image picker: Upload up to 10 photos (max 5MB each)
- Drag to reorder photos

****Step 4: Review & Submit****

- Summary of all entered data
- Button: "Create Venue"
- API: POST /api/v1/vendor/venues/

****Success Screen:****

- Checkmark animation
- Message: "Venue created successfully!"
- Button: "Add Court"

CourtRegistrationScreen

****Route:**** `/vendor/setup/court`

****Flow:****

****Step 1: Court Details****

- TextInput: Court Name
- Dropdown: Court Type (Cricket, Badminton, Tennis, Basketball, Volleyball, Football, Table Tennis, Swimming, Gym, Yoga, Other)
- TextInput: Base Price per Hour (currency input)
- TextInput: Capacity (number of players)
- Toggle: "Lights Available"
- Checkbox group: Equipment Available (Stumps, Bats, Balls, Rackets, Shuttles, etc.)
- Dropdown: Surface Type (optional)

****Step 2: Images & Details****

- Image picker: Upload up to 5 photos
- Drag to reorder

****Step 3: Availability****

- Weekly schedule (repeating):
 - Toggle for each day: Monday-Sunday
 - Time range pickers: Start time, End time
 - Example: "Monday-Friday: 06:00 - 22:00, Sat-Sun: 07:00 - 20:00"

****Step 4: Review & Submit****

- Summary
- Button: "Create Court"
- API: POST /api/v1/vendor/courts/

****Success Screen:****

- "Court created successfully!"
- Button: "Generate Time Slots"

TimeSlotGenerationScreen

****Route:**** `/vendor/setup/slots`

****Components:****

- Venue selector (dropdown)
- Court selector (dropdown)
- Date picker: "Generate for date"
- Slot duration toggle: 30 min / 60 min (radio)
- Time range: Start time – End time
- Button: "Generate Slots"
- API: POST /api/v1/vendor/courts/{court_id}/slots/generate/

****Success:****

- "16 slots generated for 2025-01-20"
- Button: "View Calendar"

4.4 Pricing Management Screens

PricingManagementScreen

****Route:**** `/vendor/pricing`

****Components:****

****Court/Service Selector:****

- Dropdown or card selector (list of courts/services)

****Current Pricing Card:****

- Base price: ₹1500/hour
- Edit button

****Time-Based Pricing (Optional):****

- Time slots:
 - Morning (06:00–12:00): ₹1200
 - Afternoon (12:00–18:00): ₹1500
 - Evening (18:00–22:00): ₹1800
- Toggle: "Use time-based pricing"
- Add/Edit buttons per slot

****Special Pricing (Optional):****

- Checkbox: "Weekend pricing"
 - Weekend price: ₹2000
- Checkbox: "Holiday pricing"
 - Holiday price: ₹2500

```
**Edit Price Modal:**
- TextInput: Price
- Save / Cancel buttons
- API: PUT /api/v1/vendor/courts/{court_id}/pricing/
```

4.5 Membership Management Screens

MembershipsScreen

```
**Route:** `/vendor/memberships`
**Components:**
```

Active Memberships List:

- Card per membership:
 - Name: "Monthly Court Pass"
 - Description: "Unlimited access for 30 days"
 - Price: ₹10,000
 - Current members: 28
 - Toggle: Active / Inactive
 - Edit / Delete buttons

```
**Button: "+ Create Membership"**
```

```
→ MembershipCreationScreen
```

MembershipCreationScreen

```
**Route:** `/vendor/memberships/create`
**Flow:**
```

- TextInput: Membership Name
- TextInput: Description
- Radio: Type (Session-Based / Credit-Based / Unlimited / Time-Based)
 - If Session-Based:
 - TextInput: Sessions Count (e.g., 12)
 - If Credit-Based:
 - TextInput: Credits Count
 - If Unlimited:
 - (no additional input)
 - If Time-Based:
 - TextInput: Validity Days
- TextInput: Price
- TextInput: Discount % (vs. pay-per-use)
- TextInput: Renewal Reminder Days (before expiry)
- Toggle: Auto-Renewal
- Button: "Create Membership"
- API: POST /api/v1/vendor/memberships/

4.6 Customer Management Screens

CustomersScreen

****Route:**** `/vendor/customers`

****Components:****

****Search & Filter:****

- Search bar: "Search by name, phone, email"
- Filter (dropdown): All / Regular / VIP / New / Inactive
- Sort (dropdown): By name / By last booking / By total spending

****Customers List:****

- FlatList
- Each item:
 - Name + avatar
 - Phone (last 4 digits masked for privacy)
 - Last booking date
 - Total bookings (count)
 - Total spending (₹)
 - Membership status (if active)
 - Tap → CustomerDetailScreen

****Empty State:****

- "No customers yet"
- "Create your first booking to add a customer"

CustomerDetailScreen

****Route:**** `/vendor/customers/{customer\_id}`

****Components:****

****Customer Info Card:****

- Name, phone, email, address
- Member since (date)
- Total bookings (count)
- Total spent (₹)
- Avg rating (stars, if customer left reviews)

****Booking History:****

- List of bookings/orders
- Tap → BookingDetailScreen

****Memberships:****

- If active, show:
 - Membership name
 - Credits/sessions used vs. total
 - Expiry date
 - Button: "Renew" / "Cancel"

****Actions:****

- Button: "+ New Booking"
- Button: "Send Message" (in-app)
- Button: "Add Membership"
- Button: "Add Note"

4.7 Offers & Promotions Screens

OffersScreen

Route: `/vendor/offers`

Components:

Active Offers:

- List of offers vendor can apply (created by Super Admin/Admin)
- Card per offer:
 - Name: "Welcome 20% Off"
 - Type: "INITIAL" / "FESTIVE"
 - Discount: "20% OFF" or "₹500 OFF"
 - Validity dates
 - Usage: "1,250 / 5,000 redeemed"
 - Status: Active / Expired / Upcoming
 - Toggle: Can only view, not edit (vendor cannot create offers)

Offer Detail Modal:

- Full offer info
- Redemption graph (optional)
- Revenue impact (₹)

4.8 Financial & Reports Screens

FinancialSummaryScreen

Route: `/vendor/financial`

Components:

Period Selector:

- Radio: This Month / Last Month / Custom Date Range

Summary Cards:

- Total Revenue (this period): ₹1,200,000
- Commission Paid: -₹240,000 (20%)
- Net Revenue: ₹960,000
- Pending Payout: ₹50,000
- Next Payout Date: 2025-02-07

Breakdown by Service:

- Venues: ₹1,000,000 (Commission: ₹200,000)
- Coaching: ₹150,000 (Commission: ₹30,000)
- Ecommerce: ₹50,000 (Commission: ₹10,000)

Daily Breakdown:

- List or chart showing daily revenue/payout
- Tap day → Daily detail

Payout History:

- List of past payouts:
 - Date, amount, status (Processed / Pending / Failed)
 - Bank account (last 4 digits)

Button: "Download Statement" (PDF/CSV)

AnalyticsScreen

****Route:**** `/vendor/analytics`

****Components:****

****Metric Selector:****

– Radio: Bookings / Revenue / Customers / Ratings

****Charts:****

– Line chart: Trend (last 30 days)

– Bar chart: By service type

– Pie chart: Category distribution

****Key Metrics:****

– Total: X (large number)

– Daily avg: X

– Trend: ↑ +15% or ↓ -5%

– Top performer: "Cricket Court A – 150 bookings"

PART 5: SECURITY REQUIREMENTS

5.1 Authentication & Authorization

– ****JWT with expiry:**** Access token 1 hour, refresh token 30 days

– ****2FA:**** TOTP (Google Authenticator, Authy) or SMS OTP

– ****Password:**** Minimum 12 chars, uppercase, lowercase, digit, special char; hash with bcrypt (rounds: 12)

– ****Session:**** Only one active session per user (device lock optional)

– ****Rate limiting:**** 5 failed login attempts → lock 30 minutes

– ****Token revocation:**** On logout, invalidate tokens

5.2 API Security

– ****HTTPS only:**** All endpoints

– ****CORS:**** Configure allowed origins (no wildcard *)

– ****CSRF Protection:**** Include CSRF token in POST/PUT/DELETE requests

– ****Input validation:**** Server-side validation for all inputs (email regex, phone format, length, special chars)

– ****SQL injection:**** Use parameterized queries (Django ORM handles this)

– ****XSS protection:**** Sanitize outputs, use Content Security Policy headers

– ****Rate limiting:**** 100 requests/minute per IP, per user

5.3 Role-Based Access Control (RBAC)

– ****Super Admin:**** Full access (all endpoints)

– ****Admin:**** Limited (vendors, customers, some analytics; no Super Admin endpoints)

– ****Vendor:**** Own data only (own bookings, customers, financial; no

other vendor data)

- ****Customer:**** Own data only
- ****Enforce at API level:**** Middleware checks role and permissions before processing

5.4 Data Encryption

- ****At rest:**** Database encryption (PostgreSQL pgcrypto), sensitive fields encrypted (PAN, bank account, 2FA secret)
- ****In transit:**** TLS 1.2+ for all API calls
- ****Mobile storage:**** Keychain (iOS), Keystore (Android) for tokens and PII

5.5 Audit & Logging

- ****Log all:**** Create, Update, Delete, Login, Permission changes
- ****Retention:**** 1 year
- ****Not logged:**** Passwords, tokens, PII (only hashed/masked)
- ****Access logs:**** IP, user agent, timestamp

5.6 Compliance

- ****PII protection:**** GDPR-compliant data export, right to be forgotten
- ****GST/Tax:**** Comply with Indian GST, invoicing requirements
- ****Payment:**** PCI DSS for payment data (use Razorpay for handling, don't store raw card data)
- ****Privacy policy & Terms:**** Linked, agreed during registration

PART 6: IMPLEMENTATION SPRINT (Suggested 3-Week Plan)

Week 1: Backend Setup & Auth

- **Day 1-2:**** Django project setup, models (User, Vendor, AdminHierarchy, CommissionConfig)
- **Day 3-4:**** Auth APIs (register, login, OTP, 2FA, token refresh)
- **Day 5:**** RBAC middleware, audit logging
- **Deliverable:**** Users can register, login, 2FA working

Week 2: Vendor & Booking Flows

- **Day 1-2:**** Court, TimeSlot, Booking models + APIs (create, list, detail)
- **Day 3-4:**** Offer model, commission logic, payment webhook
- **Day 5:**** Coach, ecommerce models + basic APIs
- **Deliverable:**** Vendors can create bookings, payments working

Week 3: Mobile App (Expo)

- **Day 1-2:**** Login, splash, navigation setup
- **Day 3-4:**** Dashboard, calendar, booking creation UI
- **Day 5:**** Connect to APIs, test end-to-end

